

MBA24155_MLWBA.R

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```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
library(fastDummies)
setwd("~/Desktop/MLWBA")
dataset = read.csv("DataMLWBA_Cleaned.csv")
dataset$Amount.Requested = as.numeric(dataset$Amount.Requested)
dataset$Amount.Funded.By.Investors = as.numeric(dataset$Amount.Funded.By.Investor
s)
```

```
## Warning: NAs introduced by coercion
```

```
dataset$Revolving.CREDIT.Balance = as.numeric(dataset$Revolving.CREDIT.Balance)
```

```
## Warning: NAs introduced by coercion
```

```
dataset$Open.CREDIT.Lines = as.numeric(dataset$Open.CREDIT.Lines)
```

```
## Warning: NAs introduced by coercion
```

```
summary(dataset)
```

```
##          ID          Amount.Requested Amount.Funded.By.Investors
## Min.      :    10      Min.      : 1000      Min.      :   -0.01
## 1st Qu.: 26323      1st Qu.: 6000      1st Qu.: 6000.00
## Median : 51087      Median :10000      Median :10000.00
## Mean    : 51931      Mean    :12504      Mean    :12096.25
## 3rd Qu.: 78336      3rd Qu.:17000      3rd Qu.:16350.00
## Max.    :104202      Max.    :35000      Max.    :35000.00
## NA's    :1                      NA's    :2
## Interest.Rate      Loan.Length      Loan.Purpose      Debt.To.Income.Ratio
## Length:2182      Length:2182      Length:2182      Length:2182
## Class :character      Class :character      Class :character      Class :character
## Mode  :character      Mode  :character      Mode  :character      Mode  :character
##
##
##
##
##          State          Home.Ownership      Monthly.Income      FICO.Range
## Length:2182      Length:2182      Min.      :   588.5      Length:2182
## Class :character      Class :character      1st Qu.:   3417.0      Class :character
## Mode  :character      Mode  :character      Median :   5000.0      Mode  :character
##                                     Mean    :   5719.9
##                                     3rd Qu.:   6833.3
##                                     Max.    :102750.0
##
## Open.CREDIT.Lines Revolving.CREDIT.Balance Inquiries.in.the.Last.6.Months
## Min.      : 2.00      Min.      :    0      Min.      :0.0000
## 1st Qu.: 7.00      1st Qu.:   5580      1st Qu.:0.0000
## Median : 9.00      Median : 11002      Median :0.0000
## Mean    :10.08      Mean    : 15329      Mean    :0.8983
## 3rd Qu.:13.00      3rd Qu.: 18844      3rd Qu.:1.0000
## Max.    :38.00      Max.    :270800      Max.    :9.0000
## NA's    :2          NA's    :2
## Employment.Length
## Length:2182
## Class :character
## Mode  :character
##
##
##
##
```

```
View(dataset)
dataset = dummy_cols(dataset,select_columns = c("Loan.Length","Loan.Purpose","Home.Ownership","Employment.Length"))
cr = cor(dataset[apply(dataset,is.numeric)])
cr
```

```
##          ID Amount.Requested Amount.Funded.By.Investors
## ID          1                      NA                      NA
## Amount.Requested      NA          1.000000000          NA
## Amount.Funded.By.Investors      NA                      NA          1
## Monthly.Income      NA          0.385857912          NA
```

##	Open.CREDIT.Lines	NA	NA	NA
##	Revolving.CREDIT.Balance	NA	NA	NA
##	Inquiries.in.the.Last.6.Months	NA	-0.011891921	NA
##	Loan.Length_36 months	NA	-0.408701706	NA
##	Loan.Length_60 months	NA	0.408701706	NA
##	Loan.Purpose_car	NA	NA	NA
##	Loan.Purpose_credit_card	NA	NA	NA
##	Loan.Purpose_debt_consolidation	NA	NA	NA
##	Loan.Purpose_educational	NA	NA	NA
##	Loan.Purpose_home_improvement	NA	NA	NA
##	Loan.Purpose_house	NA	NA	NA
##	Loan.Purpose_major_purchase	NA	NA	NA
##	Loan.Purpose_medical	NA	NA	NA
##	Loan.Purpose_moving	NA	NA	NA
##	Loan.Purpose_other	NA	NA	NA
##	Loan.Purpose_renewable_energy	NA	NA	NA
##	Loan.Purpose_small_business	NA	NA	NA
##	Loan.Purpose_vacation	NA	NA	NA
##	Loan.Purpose_wedding	NA	NA	NA
##	Loan.Purpose_NA	NA	-0.006839871	NA
##	Home.Ownership_MORTGAGE	NA	0.149417598	NA
##	Home.Ownership_OTHER	NA	-0.007368641	NA
##	Home.Ownership_OWN	NA	-0.044247855	NA
##	Home.Ownership_RENT	NA	-0.124718295	NA
##	Employment.Length_< 1 year	NA	-0.065583608	NA
##	Employment.Length_0	NA	-0.062383502	NA
##	Employment.Length_1 year	NA	-0.027472289	NA
##	Employment.Length_2 years	NA	-0.042194989	NA
##	Employment.Length_3 years	NA	-0.048155568	NA
##	Employment.Length_4 years	NA	-0.011964938	NA
##	Employment.Length_5 years	NA	-0.012581984	NA
##	Employment.Length_6 years	NA	-0.025704849	NA
##	Employment.Length_7 years	NA	0.008242044	NA
##	Employment.Length_8 years	NA	0.008558836	NA
##	Employment.Length_9 years	NA	0.050315258	NA
##	Employment.Length_10+ years	NA	0.148552254	NA
##		Monthly.Income	Open.CREDIT.Lines	
##	ID	NA	NA	
##	Amount.Requested	0.385857912	NA	
##	Amount.Funded.By.Investors	NA	NA	
##	Monthly.Income	1.000000000	NA	
##	Open.CREDIT.Lines	NA	1	
##	Revolving.CREDIT.Balance	NA	NA	
##	Inquiries.in.the.Last.6.Months	0.039143721	NA	
##	Loan.Length_36 months	-0.069451283	NA	
##	Loan.Length_60 months	0.069451283	NA	
##	Loan.Purpose_car	NA	NA	
##	Loan.Purpose_credit_card	NA	NA	
##	Loan.Purpose_debt_consolidation	NA	NA	
##	Loan.Purpose_educational	NA	NA	
##	Loan.Purpose_home_improvement	NA	NA	
##	Loan.Purpose_house	NA	NA	
##	Loan.Purpose_major_purchase	NA	NA	
##	Loan.Purpose_medical	NA	NA	

## Loan.Purpose_moving	NA	NA
## Loan.Purpose_other	NA	NA
## Loan.Purpose_renewable_energy	NA	NA
## Loan.Purpose_small_business	NA	NA
## Loan.Purpose_vacation	NA	NA
## Loan.Purpose_wedding	NA	NA
## Loan.Purpose_NA	-0.001592125	NA
## Home.Ownership_MORTGAGE	0.214345156	NA
## Home.Ownership_OTHER	0.003204365	NA
## Home.Ownership_OWN	-0.067691782	NA
## Home.Ownership_RENT	-0.177784520	NA
## Employment.Length_< 1 year	-0.054748945	NA
## Employment.Length_0	-0.066752345	NA
## Employment.Length_1 year	-0.003317116	NA
## Employment.Length_2 years	-0.014356579	NA
## Employment.Length_3 years	-0.062389796	NA
## Employment.Length_4 years	-0.017783171	NA
## Employment.Length_5 years	0.010199813	NA
## Employment.Length_6 years	-0.001467349	NA
## Employment.Length_7 years	0.050584270	NA
## Employment.Length_8 years	0.011149436	NA
## Employment.Length_9 years	-0.002684880	NA
## Employment.Length_10+ years	0.093489518	NA
##	Revolving.CREDIT.Balance	
## ID		NA
## Amount.Requested		NA
## Amount.Funded.By.Investors		NA
## Monthly.Income		NA
## Open.CREDIT.Lines		NA
## Revolving.CREDIT.Balance		1
## Inquiries.in.the.Last.6.Months		NA
## Loan.Length_36 months		NA
## Loan.Length_60 months		NA
## Loan.Purpose_car		NA
## Loan.Purpose_credit_card		NA
## Loan.Purpose_debt_consolidation		NA
## Loan.Purpose_educational		NA
## Loan.Purpose_home_improvement		NA
## Loan.Purpose_house		NA
## Loan.Purpose_major_purchase		NA
## Loan.Purpose_medical		NA
## Loan.Purpose_moving		NA
## Loan.Purpose_other		NA
## Loan.Purpose_renewable_energy		NA
## Loan.Purpose_small_business		NA
## Loan.Purpose_vacation		NA
## Loan.Purpose_wedding		NA
## Loan.Purpose_NA		NA
## Home.Ownership_MORTGAGE		NA
## Home.Ownership_OTHER		NA
## Home.Ownership_OWN		NA
## Home.Ownership_RENT		NA
## Employment.Length_< 1 year		NA
## Employment.Length_0		NA

## Employment.Length_1 year	NA		
## Employment.Length_2 years	NA		
## Employment.Length_3 years	NA		
## Employment.Length_4 years	NA		
## Employment.Length_5 years	NA		
## Employment.Length_6 years	NA		
## Employment.Length_7 years	NA		
## Employment.Length_8 years	NA		
## Employment.Length_9 years	NA		
## Employment.Length_10+ years	NA		
##	Inquiries.in.the.Last.6.Months		
## ID	NA		
## Amount.Requested	-0.011891921		
## Amount.Funded.By.Investors	NA		
## Monthly.Income	0.039143721		
## Open.CREDIT.Lines	NA		
## Revolving.CREDIT.Balance	NA		
## Inquiries.in.the.Last.6.Months	1.000000000		
## Loan.Length_36 months	-0.044088470		
## Loan.Length_60 months	0.044088470		
## Loan.Purpose_car	NA		
## Loan.Purpose_credit_card	NA		
## Loan.Purpose_debt_consolidation	NA		
## Loan.Purpose_educational	NA		
## Loan.Purpose_home_improvement	NA		
## Loan.Purpose_house	NA		
## Loan.Purpose_major_purchase	NA		
## Loan.Purpose_medical	NA		
## Loan.Purpose_moving	NA		
## Loan.Purpose_other	NA		
## Loan.Purpose_renewable_energy	NA		
## Loan.Purpose_small_business	NA		
## Loan.Purpose_vacation	NA		
## Loan.Purpose_wedding	NA		
## Loan.Purpose_NA	0.001789646		
## Home.Ownership_MORTGAGE	0.080536421		
## Home.Ownership_OTHER	-0.005219361		
## Home.Ownership_OWN	-0.023431013		
## Home.Ownership_RENT	-0.067345820		
## Employment.Length_< 1 year	0.028339412		
## Employment.Length_0	-0.006149017		
## Employment.Length_1 year	-0.002796131		
## Employment.Length_2 years	0.017618392		
## Employment.Length_3 years	-0.016169459		
## Employment.Length_4 years	-0.031371345		
## Employment.Length_5 years	0.011376561		
## Employment.Length_6 years	0.029219357		
## Employment.Length_7 years	-0.019375495		
## Employment.Length_8 years	0.010316480		
## Employment.Length_9 years	-0.016358220		
## Employment.Length_10+ years	-0.010197223		
##	Loan.Length_36 months	Loan.Length_60 months	
## ID	NA		NA
## Amount.Requested	-0.4087017057		0.4087017057

## Amount.Funded.By.Investors	NA	NA
## Monthly.Income	-0.0694512832	0.0694512832
## Open.CREDIT.Lines	NA	NA
## Revolving.CREDIT.Balance	NA	NA
## Inquiries.in.the.Last.6.Months	-0.0440884702	0.0440884702
## Loan.Length_36 months	1.0000000000	-1.0000000000
## Loan.Length_60 months	-1.0000000000	1.0000000000
## Loan.Purpose_car	NA	NA
## Loan.Purpose_credit_card	NA	NA
## Loan.Purpose_debt_consolidation	NA	NA
## Loan.Purpose_educational	NA	NA
## Loan.Purpose_home_improvement	NA	NA
## Loan.Purpose_house	NA	NA
## Loan.Purpose_major_purchase	NA	NA
## Loan.Purpose_medical	NA	NA
## Loan.Purpose_moving	NA	NA
## Loan.Purpose_other	NA	NA
## Loan.Purpose_renewable_energy	NA	NA
## Loan.Purpose_small_business	NA	NA
## Loan.Purpose_vacation	NA	NA
## Loan.Purpose_wedding	NA	NA
## Loan.Purpose_NA	0.0112954155	-0.0112954155
## Home.Ownership_MORTGAGE	-0.0742848393	0.0742848393
## Home.Ownership_OTHER	-0.0033552633	0.0033552633
## Home.Ownership_OWN	0.0387697772	-0.0387697772
## Home.Ownership_RENT	0.0534125461	-0.0534125461
## Employment.Length_< 1 year	0.0693118401	-0.0693118401
## Employment.Length_0	0.0393153728	-0.0393153728
## Employment.Length_1 year	0.0659687870	-0.0659687870
## Employment.Length_2 years	-0.0185143112	0.0185143112
## Employment.Length_3 years	-0.0269232179	0.0269232179
## Employment.Length_4 years	-0.0220239376	0.0220239376
## Employment.Length_5 years	-0.0234763063	0.0234763063
## Employment.Length_6 years	0.0007886095	-0.0007886095
## Employment.Length_7 years	0.0375048608	-0.0375048608
## Employment.Length_8 years	-0.0138772478	0.0138772478
## Employment.Length_9 years	-0.0251507136	0.0251507136
## Employment.Length_10+ years	-0.0472434296	0.0472434296
##	Loan.Purpose_car	Loan.Purpose_credit_card
## ID	NA	NA
## Amount.Requested	NA	NA
## Amount.Funded.By.Investors	NA	NA
## Monthly.Income	NA	NA
## Open.CREDIT.Lines	NA	NA
## Revolving.CREDIT.Balance	NA	NA
## Inquiries.in.the.Last.6.Months	NA	NA
## Loan.Length_36 months	NA	NA
## Loan.Length_60 months	NA	NA
## Loan.Purpose_car	1	NA
## Loan.Purpose_credit_card	NA	1
## Loan.Purpose_debt_consolidation	NA	NA
## Loan.Purpose_educational	NA	NA
## Loan.Purpose_home_improvement	NA	NA
## Loan.Purpose_house	NA	NA

## Loan.Purpose_major_purchase	NA	NA
## Loan.Purpose_medical	NA	NA
## Loan.Purpose_moving	NA	NA
## Loan.Purpose_other	NA	NA
## Loan.Purpose_renewable_energy	NA	NA
## Loan.Purpose_small_business	NA	NA
## Loan.Purpose_vacation	NA	NA
## Loan.Purpose_wedding	NA	NA
## Loan.Purpose_NA	NA	NA
## Home.Ownership_MORTGAGE	NA	NA
## Home.Ownership_OTHER	NA	NA
## Home.Ownership_OWN	NA	NA
## Home.Ownership_RENT	NA	NA
## Employment.Length_< 1 year	NA	NA
## Employment.Length_0	NA	NA
## Employment.Length_1 year	NA	NA
## Employment.Length_2 years	NA	NA
## Employment.Length_3 years	NA	NA
## Employment.Length_4 years	NA	NA
## Employment.Length_5 years	NA	NA
## Employment.Length_6 years	NA	NA
## Employment.Length_7 years	NA	NA
## Employment.Length_8 years	NA	NA
## Employment.Length_9 years	NA	NA
## Employment.Length_10+ years	NA	NA
##	Loan.Purpose_debt_consolidation	
## ID		NA
## Amount.Requested		NA
## Amount.Funded.By.Investors		NA
## Monthly.Income		NA
## Open.CREDIT.Lines		NA
## Revolving.CREDIT.Balance		NA
## Inquiries.in.the.Last.6.Months		NA
## Loan.Length_36 months		NA
## Loan.Length_60 months		NA
## Loan.Purpose_car		NA
## Loan.Purpose_credit_card		NA
## Loan.Purpose_debt_consolidation		1
## Loan.Purpose_educational		NA
## Loan.Purpose_home_improvement		NA
## Loan.Purpose_house		NA
## Loan.Purpose_major_purchase		NA
## Loan.Purpose_medical		NA
## Loan.Purpose_moving		NA
## Loan.Purpose_other		NA
## Loan.Purpose_renewable_energy		NA
## Loan.Purpose_small_business		NA
## Loan.Purpose_vacation		NA
## Loan.Purpose_wedding		NA
## Loan.Purpose_NA		NA
## Home.Ownership_MORTGAGE		NA
## Home.Ownership_OTHER		NA
## Home.Ownership_OWN		NA
## Home.Ownership_RENT		NA

## Employment.Length_< 1 year	NA
## Employment.Length_0	NA
## Employment.Length_1 year	NA
## Employment.Length_2 years	NA
## Employment.Length_3 years	NA
## Employment.Length_4 years	NA
## Employment.Length_5 years	NA
## Employment.Length_6 years	NA
## Employment.Length_7 years	NA
## Employment.Length_8 years	NA
## Employment.Length_9 years	NA
## Employment.Length_10+ years	NA
##	Loan.Purpose_educational
## ID	NA
## Amount.Requested	NA
## Amount.Funded.By.Investors	NA
## Monthly.Income	NA
## Open.CREDIT.Lines	NA
## Revolving.CREDIT.Balance	NA
## Inquiries.in.the.Last.6.Months	NA
## Loan.Length_36 months	NA
## Loan.Length_60 months	NA
## Loan.Purpose_car	NA
## Loan.Purpose_credit_card	NA
## Loan.Purpose_debt_consolidation	NA
## Loan.Purpose_educational	1
## Loan.Purpose_home_improvement	NA
## Loan.Purpose_house	NA
## Loan.Purpose_major_purchase	NA
## Loan.Purpose_medical	NA
## Loan.Purpose_moving	NA
## Loan.Purpose_other	NA
## Loan.Purpose_renewable_energy	NA
## Loan.Purpose_small_business	NA
## Loan.Purpose_vacation	NA
## Loan.Purpose_wedding	NA
## Loan.Purpose_NA	NA
## Home.Ownership_MORTGAGE	NA
## Home.Ownership_OTHER	NA
## Home.Ownership_OWN	NA
## Home.Ownership_RENT	NA
## Employment.Length_< 1 year	NA
## Employment.Length_0	NA
## Employment.Length_1 year	NA
## Employment.Length_2 years	NA
## Employment.Length_3 years	NA
## Employment.Length_4 years	NA
## Employment.Length_5 years	NA
## Employment.Length_6 years	NA
## Employment.Length_7 years	NA
## Employment.Length_8 years	NA
## Employment.Length_9 years	NA
## Employment.Length_10+ years	NA
##	Loan.Purpose_home_improvement

## ID	NA
## Amount.Requested	NA
## Amount.Funded.By.Investors	NA
## Monthly.Income	NA
## Open.CREDIT.Lines	NA
## Revolving.CREDIT.Balance	NA
## Inquiries.in.the.Last.6.Months	NA
## Loan.Length_36 months	NA
## Loan.Length_60 months	NA
## Loan.Purpose_car	NA
## Loan.Purpose_credit_card	NA
## Loan.Purpose_debt_consolidation	NA
## Loan.Purpose_educational	NA
## Loan.Purpose_home_improvement	1
## Loan.Purpose_house	NA
## Loan.Purpose_major_purchase	NA
## Loan.Purpose_medical	NA
## Loan.Purpose_moving	NA
## Loan.Purpose_other	NA
## Loan.Purpose_renewable_energy	NA
## Loan.Purpose_small_business	NA
## Loan.Purpose_vacation	NA
## Loan.Purpose_wedding	NA
## Loan.Purpose_NA	NA
## Home.Ownership_MORTGAGE	NA
## Home.Ownership_OTHER	NA
## Home.Ownership_OWN	NA
## Home.Ownership_RENT	NA
## Employment.Length_< 1 year	NA
## Employment.Length_0	NA
## Employment.Length_1 year	NA
## Employment.Length_2 years	NA
## Employment.Length_3 years	NA
## Employment.Length_4 years	NA
## Employment.Length_5 years	NA
## Employment.Length_6 years	NA
## Employment.Length_7 years	NA
## Employment.Length_8 years	NA
## Employment.Length_9 years	NA
## Employment.Length_10+ years	NA
##	Loan.Purpose_house Loan.Purpose_major_purchase
## ID	NA NA
## Amount.Requested	NA NA
## Amount.Funded.By.Investors	NA NA
## Monthly.Income	NA NA
## Open.CREDIT.Lines	NA NA
## Revolving.CREDIT.Balance	NA NA
## Inquiries.in.the.Last.6.Months	NA NA
## Loan.Length_36 months	NA NA
## Loan.Length_60 months	NA NA
## Loan.Purpose_car	NA NA
## Loan.Purpose_credit_card	NA NA
## Loan.Purpose_debt_consolidation	NA NA
## Loan.Purpose_educational	NA NA

## Loan.Purpose_home_improvement	NA	NA
## Loan.Purpose_house	1	NA
## Loan.Purpose_major_purchase	NA	1
## Loan.Purpose_medical	NA	NA
## Loan.Purpose_moving	NA	NA
## Loan.Purpose_other	NA	NA
## Loan.Purpose_renewable_energy	NA	NA
## Loan.Purpose_small_business	NA	NA
## Loan.Purpose_vacation	NA	NA
## Loan.Purpose_wedding	NA	NA
## Loan.Purpose_NA	NA	NA
## Home.Ownership_MORTGAGE	NA	NA
## Home.Ownership_OTHER	NA	NA
## Home.Ownership_OWN	NA	NA
## Home.Ownership_RENT	NA	NA
## Employment.Length_< 1 year	NA	NA
## Employment.Length_0	NA	NA
## Employment.Length_1 year	NA	NA
## Employment.Length_2 years	NA	NA
## Employment.Length_3 years	NA	NA
## Employment.Length_4 years	NA	NA
## Employment.Length_5 years	NA	NA
## Employment.Length_6 years	NA	NA
## Employment.Length_7 years	NA	NA
## Employment.Length_8 years	NA	NA
## Employment.Length_9 years	NA	NA
## Employment.Length_10+ years	NA	NA
##	Loan.Purpose_medical	Loan.Purpose_moving
## ID	NA	NA
## Amount.Requested	NA	NA
## Amount.Funded.By.Investors	NA	NA
## Monthly.Income	NA	NA
## Open.CREDIT.Lines	NA	NA
## Revolving.CREDIT.Balance	NA	NA
## Inquiries.in.the.Last.6.Months	NA	NA
## Loan.Length_36 months	NA	NA
## Loan.Length_60 months	NA	NA
## Loan.Purpose_car	NA	NA
## Loan.Purpose_credit_card	NA	NA
## Loan.Purpose_debt_consolidation	NA	NA
## Loan.Purpose_educational	NA	NA
## Loan.Purpose_home_improvement	NA	NA
## Loan.Purpose_house	NA	NA
## Loan.Purpose_major_purchase	NA	NA
## Loan.Purpose_medical	1	NA
## Loan.Purpose_moving	NA	1
## Loan.Purpose_other	NA	NA
## Loan.Purpose_renewable_energy	NA	NA
## Loan.Purpose_small_business	NA	NA
## Loan.Purpose_vacation	NA	NA
## Loan.Purpose_wedding	NA	NA
## Loan.Purpose_NA	NA	NA
## Home.Ownership_MORTGAGE	NA	NA
## Home.Ownership_OTHER	NA	NA

## Home.Ownership_OWN	NA	NA
## Home.Ownership_RENT	NA	NA
## Employment.Length_< 1 year	NA	NA
## Employment.Length_0	NA	NA
## Employment.Length_1 year	NA	NA
## Employment.Length_2 years	NA	NA
## Employment.Length_3 years	NA	NA
## Employment.Length_4 years	NA	NA
## Employment.Length_5 years	NA	NA
## Employment.Length_6 years	NA	NA
## Employment.Length_7 years	NA	NA
## Employment.Length_8 years	NA	NA
## Employment.Length_9 years	NA	NA
## Employment.Length_10+ years	NA	NA
##	Loan.Purpose_other	
## ID	NA	
## Amount.Requested	NA	
## Amount.Funded.By.Investors	NA	
## Monthly.Income	NA	
## Open.CREDIT.Lines	NA	
## Revolving.CREDIT.Balance	NA	
## Inquiries.in.the.Last.6.Months	NA	
## Loan.Length_36 months	NA	
## Loan.Length_60 months	NA	
## Loan.Purpose_car	NA	
## Loan.Purpose_credit_card	NA	
## Loan.Purpose_debt_consolidation	NA	
## Loan.Purpose_educational	NA	
## Loan.Purpose_home_improvement	NA	
## Loan.Purpose_house	NA	
## Loan.Purpose_major_purchase	NA	
## Loan.Purpose_medical	NA	
## Loan.Purpose_moving	NA	
## Loan.Purpose_other	1	
## Loan.Purpose_renewable_energy	NA	
## Loan.Purpose_small_business	NA	
## Loan.Purpose_vacation	NA	
## Loan.Purpose_wedding	NA	
## Loan.Purpose_NA	NA	
## Home.Ownership_MORTGAGE	NA	
## Home.Ownership_OTHER	NA	
## Home.Ownership_OWN	NA	
## Home.Ownership_RENT	NA	
## Employment.Length_< 1 year	NA	
## Employment.Length_0	NA	
## Employment.Length_1 year	NA	
## Employment.Length_2 years	NA	
## Employment.Length_3 years	NA	
## Employment.Length_4 years	NA	
## Employment.Length_5 years	NA	
## Employment.Length_6 years	NA	
## Employment.Length_7 years	NA	
## Employment.Length_8 years	NA	
## Employment.Length_9 years	NA	

## Employment.Length_10+ years	NA
##	Loan.Purpose_renewable_energy
## ID	NA
## Amount.Requested	NA
## Amount.Funded.By.Investors	NA
## Monthly.Income	NA
## Open.CREDIT.Lines	NA
## Revolving.CREDIT.Balance	NA
## Inquiries.in.the.Last.6.Months	NA
## Loan.Length_36 months	NA
## Loan.Length_60 months	NA
## Loan.Purpose_car	NA
## Loan.Purpose_credit_card	NA
## Loan.Purpose_debt_consolidation	NA
## Loan.Purpose_educational	NA
## Loan.Purpose_home_improvement	NA
## Loan.Purpose_house	NA
## Loan.Purpose_major_purchase	NA
## Loan.Purpose_medical	NA
## Loan.Purpose_moving	NA
## Loan.Purpose_other	NA
## Loan.Purpose_renewable_energy	1
## Loan.Purpose_small_business	NA
## Loan.Purpose_vacation	NA
## Loan.Purpose_wedding	NA
## Loan.Purpose_NA	NA
## Home.Ownership_MORTGAGE	NA
## Home.Ownership_OTHER	NA
## Home.Ownership_OWN	NA
## Home.Ownership_RENT	NA
## Employment.Length_< 1 year	NA
## Employment.Length_0	NA
## Employment.Length_1 year	NA
## Employment.Length_2 years	NA
## Employment.Length_3 years	NA
## Employment.Length_4 years	NA
## Employment.Length_5 years	NA
## Employment.Length_6 years	NA
## Employment.Length_7 years	NA
## Employment.Length_8 years	NA
## Employment.Length_9 years	NA
## Employment.Length_10+ years	NA
##	Loan.Purpose_small_business
## ID	NA
## Amount.Requested	NA
## Amount.Funded.By.Investors	NA
## Monthly.Income	NA
## Open.CREDIT.Lines	NA
## Revolving.CREDIT.Balance	NA
## Inquiries.in.the.Last.6.Months	NA
## Loan.Length_36 months	NA
## Loan.Length_60 months	NA
## Loan.Purpose_car	NA
## Loan.Purpose_credit_card	NA

## Loan.Purpose_debt_consolidation	NA	
## Loan.Purpose_educational	NA	
## Loan.Purpose_home_improvement	NA	
## Loan.Purpose_house	NA	
## Loan.Purpose_major_purchase	NA	
## Loan.Purpose_medical	NA	
## Loan.Purpose_moving	NA	
## Loan.Purpose_other	NA	
## Loan.Purpose_renewable_energy	NA	
## Loan.Purpose_small_business	1	
## Loan.Purpose_vacation	NA	
## Loan.Purpose_wedding	NA	
## Loan.Purpose_NA	NA	
## Home.Ownership_MORTGAGE	NA	
## Home.Ownership_OTHER	NA	
## Home.Ownership_OWN	NA	
## Home.Ownership_RENT	NA	
## Employment.Length_< 1 year	NA	
## Employment.Length_0	NA	
## Employment.Length_1 year	NA	
## Employment.Length_2 years	NA	
## Employment.Length_3 years	NA	
## Employment.Length_4 years	NA	
## Employment.Length_5 years	NA	
## Employment.Length_6 years	NA	
## Employment.Length_7 years	NA	
## Employment.Length_8 years	NA	
## Employment.Length_9 years	NA	
## Employment.Length_10+ years	NA	
##	Loan.Purpose_vacation	Loan.Purpose_wedding
## ID	NA	NA
## Amount.Requested	NA	NA
## Amount.Funded.By.Investors	NA	NA
## Monthly.Income	NA	NA
## Open.CREDIT.Lines	NA	NA
## Revolving.CREDIT.Balance	NA	NA
## Inquiries.in.the.Last.6.Months	NA	NA
## Loan.Length_36 months	NA	NA
## Loan.Length_60 months	NA	NA
## Loan.Purpose_car	NA	NA
## Loan.Purpose_credit_card	NA	NA
## Loan.Purpose_debt_consolidation	NA	NA
## Loan.Purpose_educational	NA	NA
## Loan.Purpose_home_improvement	NA	NA
## Loan.Purpose_house	NA	NA
## Loan.Purpose_major_purchase	NA	NA
## Loan.Purpose_medical	NA	NA
## Loan.Purpose_moving	NA	NA
## Loan.Purpose_other	NA	NA
## Loan.Purpose_renewable_energy	NA	NA
## Loan.Purpose_small_business	NA	NA
## Loan.Purpose_vacation	1	NA
## Loan.Purpose_wedding	NA	1
## Loan.Purpose_NA	NA	NA

## Home.Ownership_MORTGAGE	NA	NA
## Home.Ownership_OTHER	NA	NA
## Home.Ownership_OWN	NA	NA
## Home.Ownership_RENT	NA	NA
## Employment.Length_< 1 year	NA	NA
## Employment.Length_0	NA	NA
## Employment.Length_1 year	NA	NA
## Employment.Length_2 years	NA	NA
## Employment.Length_3 years	NA	NA
## Employment.Length_4 years	NA	NA
## Employment.Length_5 years	NA	NA
## Employment.Length_6 years	NA	NA
## Employment.Length_7 years	NA	NA
## Employment.Length_8 years	NA	NA
## Employment.Length_9 years	NA	NA
## Employment.Length_10+ years	NA	NA
##	Loan.Purpose_NA	Home.Ownership_MORTGAGE
## ID	NA	NA
## Amount.Requested	-0.0068398707	0.1494175976
## Amount.Funded.By.Investors	NA	NA
## Monthly.Income	-0.0015921255	0.2143451558
## Open.CREDIT.Lines	NA	NA
## Revolving.CREDIT.Balance	NA	NA
## Inquiries.in.the.Last.6.Months	0.0017896461	0.0805364212
## Loan.Length_36 months	0.0112954155	-0.0742848393
## Loan.Length_60 months	-0.0112954155	0.0742848393
## Loan.Purpose_car	NA	NA
## Loan.Purpose_credit_card	NA	NA
## Loan.Purpose_debt_consolidation	NA	NA
## Loan.Purpose_educational	NA	NA
## Loan.Purpose_home_improvement	NA	NA
## Loan.Purpose_house	NA	NA
## Loan.Purpose_major_purchase	NA	NA
## Loan.Purpose_medical	NA	NA
## Loan.Purpose_moving	NA	NA
## Loan.Purpose_other	NA	NA
## Loan.Purpose_renewable_energy	NA	NA
## Loan.Purpose_small_business	NA	NA
## Loan.Purpose_vacation	NA	NA
## Loan.Purpose_wedding	NA	NA
## Loan.Purpose_NA	1.0000000000	0.0230874299
## Home.Ownership_MORTGAGE	0.0230874299	1.0000000000
## Home.Ownership_OTHER	-0.0009176419	-0.0397463850
## Home.Ownership_OWN	-0.0063621164	-0.2755662472
## Home.Ownership_RENT	-0.0195503494	-0.8467962624
## Employment.Length_< 1 year	-0.0072605248	-0.1059236142
## Employment.Length_0	-0.0038982908	0.0085058041
## Employment.Length_1 year	-0.0059826756	-0.0853422335
## Employment.Length_2 years	-0.0070975308	-0.0488838287
## Employment.Length_3 years	-0.0068393567	-0.0774102185
## Employment.Length_4 years	-0.0060233983	-0.0387431872
## Employment.Length_5 years	-0.0064206114	-0.0007870462
## Employment.Length_6 years	0.0837114268	-0.0266635100
## Employment.Length_7 years	-0.0048624484	0.0320139688

## Employment.Length_8 years	-0.0045432941	0.0612646941
## Employment.Length_9 years	-0.0037520483	0.0213175959
## Employment.Length_10+ years	-0.0127480116	0.1902114448
##	Home.Ownership_OTHER	Home.Ownership_OWN
## ID	NA	NA
## Amount.Requested	-0.0073686408	-0.044247855
## Amount.Funded.By.Investors	NA	NA
## Monthly.Income	0.0032043648	-0.067691782
## Open.CREDIT.Lines	NA	NA
## Revolving.CREDIT.Balance	NA	NA
## Inquiries.in.the.Last.6.Months	-0.0052193612	-0.023431013
## Loan.Length_36 months	-0.0033552633	0.038769777
## Loan.Length_60 months	0.0033552633	-0.038769777
## Loan.Purpose_car	NA	NA
## Loan.Purpose_credit_card	NA	NA
## Loan.Purpose_debt_consolidation	NA	NA
## Loan.Purpose_educational	NA	NA
## Loan.Purpose_home_improvement	NA	NA
## Loan.Purpose_house	NA	NA
## Loan.Purpose_major_purchase	NA	NA
## Loan.Purpose_medical	NA	NA
## Loan.Purpose_moving	NA	NA
## Loan.Purpose_other	NA	NA
## Loan.Purpose_renewable_energy	NA	NA
## Loan.Purpose_small_business	NA	NA
## Loan.Purpose_vacation	NA	NA
## Loan.Purpose_wedding	NA	NA
## Loan.Purpose_NA	-0.0009176419	-0.006362116
## Home.Ownership_MORTGAGE	-0.0397463850	-0.275566247
## Home.Ownership_OTHER	1.0000000000	-0.012732993
## Home.Ownership_OWN	-0.0127329931	1.0000000000
## Home.Ownership_RENT	-0.0391276183	-0.271276267
## Employment.Length_< 1 year	-0.0145310469	0.004131005
## Employment.Length_0	-0.0078019493	0.022116958
## Employment.Length_1 year	-0.0119735890	0.033572601
## Employment.Length_2 years	0.0575427322	-0.008552156
## Employment.Length_3 years	0.0232765987	-0.002234910
## Employment.Length_4 years	-0.0120550905	0.006575379
## Employment.Length_5 years	-0.0128500635	-0.009770365
## Employment.Length_6 years	-0.0109619668	0.028877526
## Employment.Length_7 years	-0.0097315921	-0.005283192
## Employment.Length_8 years	-0.0090928439	-0.013435868
## Employment.Length_9 years	-0.0075092629	-0.012566518
## Employment.Length_10+ years	-0.0011393979	-0.024130156
##	Home.Ownership_RENT	Employment.Length_< 1 year
## ID	NA	NA
## Amount.Requested	-0.124718295	-0.065583608
## Amount.Funded.By.Investors	NA	NA
## Monthly.Income	-0.177784520	-0.054748945
## Open.CREDIT.Lines	NA	NA
## Revolving.CREDIT.Balance	NA	NA
## Inquiries.in.the.Last.6.Months	-0.067345820	0.028339412
## Loan.Length_36 months	0.053412546	0.069311840
## Loan.Length_60 months	-0.053412546	-0.069311840

## Loan.Purpose_car	NA	NA
## Loan.Purpose_credit_card	NA	NA
## Loan.Purpose_debt_consolidation	NA	NA
## Loan.Purpose_educational	NA	NA
## Loan.Purpose_home_improvement	NA	NA
## Loan.Purpose_house	NA	NA
## Loan.Purpose_major_purchase	NA	NA
## Loan.Purpose_medical	NA	NA
## Loan.Purpose_moving	NA	NA
## Loan.Purpose_other	NA	NA
## Loan.Purpose_renewable_energy	NA	NA
## Loan.Purpose_small_business	NA	NA
## Loan.Purpose_vacation	NA	NA
## Loan.Purpose_wedding	NA	NA
## Loan.Purpose_NA	-0.019550349	-0.007260525
## Home.Ownership_MORTGAGE	-0.846796262	-0.105923614
## Home.Ownership_OTHER	-0.039127618	-0.014531047
## Home.Ownership_OWN	-0.271276267	0.004131005
## Home.Ownership_RENT	1.000000000	0.105044884
## Employment.Length_< 1 year	0.105044884	1.000000000
## Employment.Length_0	-0.019973262	-0.061730232
## Employment.Length_1 year	0.068074305	-0.094736893
## Employment.Length_2 years	0.048693268	-0.112390852
## Employment.Length_3 years	0.076736803	-0.108302613
## Employment.Length_4 years	0.036224018	-0.095381745
## Employment.Length_5 years	0.007249029	-0.101671695
## Employment.Length_6 years	0.011806538	-0.086732780
## Employment.Length_7 years	-0.028322902	-0.076997865
## Employment.Length_8 years	-0.053196523	-0.071943991
## Employment.Length_9 years	-0.013810099	-0.059414452
## Employment.Length_10+ years	-0.177130850	-0.201867372
##	Employment.Length_0	Employment.Length_1 year
## ID	NA	NA
## Amount.Requested	-0.062383502	-0.027472289
## Amount.Funded.By.Investors	NA	NA
## Monthly.Income	-0.066752345	-0.003317116
## Open.CREDIT.Lines	NA	NA
## Revolving.CREDIT.Balance	NA	NA
## Inquiries.in.the.Last.6.Months	-0.006149017	-0.002796131
## Loan.Length_36 months	0.039315373	0.065968787
## Loan.Length_60 months	-0.039315373	-0.065968787
## Loan.Purpose_car	NA	NA
## Loan.Purpose_credit_card	NA	NA
## Loan.Purpose_debt_consolidation	NA	NA
## Loan.Purpose_educational	NA	NA
## Loan.Purpose_home_improvement	NA	NA
## Loan.Purpose_house	NA	NA
## Loan.Purpose_major_purchase	NA	NA
## Loan.Purpose_medical	NA	NA
## Loan.Purpose_moving	NA	NA
## Loan.Purpose_other	NA	NA
## Loan.Purpose_renewable_energy	NA	NA
## Loan.Purpose_small_business	NA	NA
## Loan.Purpose_vacation	NA	NA

## Loan.Purpose_wedding	NA	NA
## Loan.Purpose_NA	-0.003898291	-0.005982676
## Home.Ownership_MORTGAGE	0.008505804	-0.085342233
## Home.Ownership_OTHER	-0.007801949	-0.011973589
## Home.Ownership_OWN	0.022116958	0.033572601
## Home.Ownership_RENT	-0.019973262	0.068074305
## Employment.Length_< 1 year	-0.061730232	-0.094736893
## Employment.Length_0	1.000000000	-0.050865738
## Employment.Length_1 year	-0.050865738	1.000000000
## Employment.Length_2 years	-0.060344429	-0.092610112
## Employment.Length_3 years	-0.058149389	-0.089241401
## Employment.Length_4 years	-0.051211970	-0.078594599
## Employment.Length_5 years	-0.054589143	-0.083777521
## Employment.Length_6 years	-0.046568204	-0.071467849
## Employment.Length_7 years	-0.041341374	-0.063446275
## Employment.Length_8 years	-0.038627868	-0.059281880
## Employment.Length_9 years	-0.031900560	-0.048957535
## Employment.Length_10+ years	-0.108385790	-0.166338804
##	Employment.Length_2 years	
## ID	NA	
## Amount.Requested	-0.042194989	
## Amount.Funded.By.Investors	NA	
## Monthly.Income	-0.014356579	
## Open.CREDIT.Lines	NA	
## Revolving.CREDIT.Balance	NA	
## Inquiries.in.the.Last.6.Months	0.017618392	
## Loan.Length_36 months	-0.018514311	
## Loan.Length_60 months	0.018514311	
## Loan.Purpose_car	NA	
## Loan.Purpose_credit_card	NA	
## Loan.Purpose_debt_consolidation	NA	
## Loan.Purpose_educational	NA	
## Loan.Purpose_home_improvement	NA	
## Loan.Purpose_house	NA	
## Loan.Purpose_major_purchase	NA	
## Loan.Purpose_medical	NA	
## Loan.Purpose_moving	NA	
## Loan.Purpose_other	NA	
## Loan.Purpose_renewable_energy	NA	
## Loan.Purpose_small_business	NA	
## Loan.Purpose_vacation	NA	
## Loan.Purpose_wedding	NA	
## Loan.Purpose_NA	-0.007097531	
## Home.Ownership_MORTGAGE	-0.048883829	
## Home.Ownership_OTHER	0.057542732	
## Home.Ownership_OWN	-0.008552156	
## Home.Ownership_RENT	0.048693268	
## Employment.Length_< 1 year	-0.112390852	
## Employment.Length_0	-0.060344429	
## Employment.Length_1 year	-0.092610112	
## Employment.Length_2 years	1.000000000	
## Employment.Length_3 years	-0.105871291	
## Employment.Length_4 years	-0.093240487	
## Employment.Length_5 years	-0.099389232	

## Employment.Length_6 years	-0.084785687
## Employment.Length_7 years	-0.075269314
## Employment.Length_8 years	-0.070328896
## Employment.Length_9 years	-0.058080637
## Employment.Length_10+ years	-0.197335583
##	Employment.Length_3 years
## ID	NA
## Amount.Requested	-0.048155568
## Amount.Funded.By.Investors	NA
## Monthly.Income	-0.062389796
## Open.CREDIT.Lines	NA
## Revolving.CREDIT.Balance	NA
## Inquiries.in.the.Last.6.Months	-0.016169459
## Loan.Length_36 months	-0.026923218
## Loan.Length_60 months	0.026923218
## Loan.Purpose_car	NA
## Loan.Purpose_credit_card	NA
## Loan.Purpose_debt_consolidation	NA
## Loan.Purpose_educational	NA
## Loan.Purpose_home_improvement	NA
## Loan.Purpose_house	NA
## Loan.Purpose_major_purchase	NA
## Loan.Purpose_medical	NA
## Loan.Purpose_moving	NA
## Loan.Purpose_other	NA
## Loan.Purpose_renewable_energy	NA
## Loan.Purpose_small_business	NA
## Loan.Purpose_vacation	NA
## Loan.Purpose_wedding	NA
## Loan.Purpose_NA	-0.006839357
## Home.Ownership_MORTGAGE	-0.077410218
## Home.Ownership_OTHER	0.023276599
## Home.Ownership_OWN	-0.002234910
## Home.Ownership_RENT	0.076736803
## Employment.Length_< 1 year	-0.108302613
## Employment.Length_0	-0.058149389
## Employment.Length_1 year	-0.089241401
## Employment.Length_2 years	-0.105871291
## Employment.Length_3 years	1.000000000
## Employment.Length_4 years	-0.089848846
## Employment.Length_5 years	-0.095773929
## Employment.Length_6 years	-0.081701591
## Employment.Length_7 years	-0.072531378
## Employment.Length_8 years	-0.067770668
## Employment.Length_9 years	-0.055967942
## Employment.Length_10+ years	-0.190157463
##	Employment.Length_4 years
## ID	NA
## Amount.Requested	-0.011964938
## Amount.Funded.By.Investors	NA
## Monthly.Income	-0.017783171
## Open.CREDIT.Lines	NA
## Revolving.CREDIT.Balance	NA
## Inquiries.in.the.Last.6.Months	-0.031371345

## Loan.Length_36 months	-0.022023938
## Loan.Length_60 months	0.022023938
## Loan.Purpose_car	NA
## Loan.Purpose_credit_card	NA
## Loan.Purpose_debt_consolidation	NA
## Loan.Purpose_educational	NA
## Loan.Purpose_home_improvement	NA
## Loan.Purpose_house	NA
## Loan.Purpose_major_purchase	NA
## Loan.Purpose_medical	NA
## Loan.Purpose_moving	NA
## Loan.Purpose_other	NA
## Loan.Purpose_renewable_energy	NA
## Loan.Purpose_small_business	NA
## Loan.Purpose_vacation	NA
## Loan.Purpose_wedding	NA
## Loan.Purpose_NA	-0.006023398
## Home.Ownership_MORTGAGE	-0.038743187
## Home.Ownership_OTHER	-0.012055090
## Home.Ownership_OWN	0.006575379
## Home.Ownership_RENT	0.036224018
## Employment.Length_< 1 year	-0.095381745
## Employment.Length_0	-0.051211970
## Employment.Length_1 year	-0.078594599
## Employment.Length_2 years	-0.093240487
## Employment.Length_3 years	-0.089848846
## Employment.Length_4 years	1.000000000
## Employment.Length_5 years	-0.084347775
## Employment.Length_6 years	-0.071954314
## Employment.Length_7 years	-0.063878139
## Employment.Length_8 years	-0.059685398
## Employment.Length_9 years	-0.049290777
## Employment.Length_10+ years	-0.167471033
##	Employment.Length_5 years
## ID	NA
## Amount.Requested	-0.0125819842
## Amount.Funded.By.Investors	NA
## Monthly.Income	0.0101998134
## Open.CREDIT.Lines	NA
## Revolving.CREDIT.Balance	NA
## Inquiries.in.the.Last.6.Months	0.0113765611
## Loan.Length_36 months	-0.0234763063
## Loan.Length_60 months	0.0234763063
## Loan.Purpose_car	NA
## Loan.Purpose_credit_card	NA
## Loan.Purpose_debt_consolidation	NA
## Loan.Purpose_educational	NA
## Loan.Purpose_home_improvement	NA
## Loan.Purpose_house	NA
## Loan.Purpose_major_purchase	NA
## Loan.Purpose_medical	NA
## Loan.Purpose_moving	NA
## Loan.Purpose_other	NA
## Loan.Purpose_renewable_energy	NA

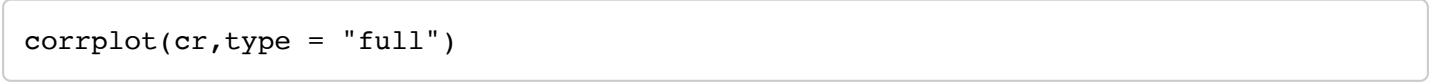
## Loan.Purpose_small_business	NA
## Loan.Purpose_vacation	NA
## Loan.Purpose_wedding	NA
## Loan.Purpose_NA	-0.0064206114
## Home.Ownership_MORTGAGE	-0.0007870462
## Home.Ownership_OTHER	-0.0128500635
## Home.Ownership_OWN	-0.0097703650
## Home.Ownership_RENT	0.0072490294
## Employment.Length_< 1 year	-0.1016716946
## Employment.Length_0	-0.0545891433
## Employment.Length_1 year	-0.0837775210
## Employment.Length_2 years	-0.0993892320
## Employment.Length_3 years	-0.0957739293
## Employment.Length_4 years	-0.0843477752
## Employment.Length_5 years	1.0000000000
## Employment.Length_6 years	-0.0766993421
## Employment.Length_7 years	-0.0680905833
## Employment.Length_8 years	-0.0636213522
## Employment.Length_9 years	-0.0525412578
## Employment.Length_10+ years	-0.1785149119
##	Employment.Length_6 years
## ID	NA
## Amount.Requested	-0.0257048492
## Amount.Funded.By.Investors	NA
## Monthly.Income	-0.0014673495
## Open.CREDIT.Lines	NA
## Revolving.CREDIT.Balance	NA
## Inquiries.in.the.Last.6.Months	0.0292193565
## Loan.Length_36 months	0.0007886095
## Loan.Length_60 months	-0.0007886095
## Loan.Purpose_car	NA
## Loan.Purpose_credit_card	NA
## Loan.Purpose_debt_consolidation	NA
## Loan.Purpose_educational	NA
## Loan.Purpose_home_improvement	NA
## Loan.Purpose_house	NA
## Loan.Purpose_major_purchase	NA
## Loan.Purpose_medical	NA
## Loan.Purpose_moving	NA
## Loan.Purpose_other	NA
## Loan.Purpose_renewable_energy	NA
## Loan.Purpose_small_business	NA
## Loan.Purpose_vacation	NA
## Loan.Purpose_wedding	NA
## Loan.Purpose_NA	0.0837114268
## Home.Ownership_MORTGAGE	-0.0266635100
## Home.Ownership_OTHER	-0.0109619668
## Home.Ownership_OWN	0.0288775262
## Home.Ownership_RENT	0.0118065383
## Employment.Length_< 1 year	-0.0867327805
## Employment.Length_0	-0.0465682037
## Employment.Length_1 year	-0.0714678492
## Employment.Length_2 years	-0.0847856867
## Employment.Length_3 years	-0.0817015908

## Employment.Length_4 years	-0.0719543143
## Employment.Length_5 years	-0.0766993421
## Employment.Length_6 years	1.0000000000
## Employment.Length_7 years	-0.0580858383
## Employment.Length_8 years	-0.0542732842
## Employment.Length_9 years	-0.0448212199
## Employment.Length_10+ years	-0.1522852031
##	Employment.Length_7 years
## ID	NA
## Amount.Requested	0.008242044
## Amount.Funded.By.Investors	NA
## Monthly.Income	0.050584270
## Open.CREDIT.Lines	NA
## Revolving.CREDIT.Balance	NA
## Inquiries.in.the.Last.6.Months	-0.019375495
## Loan.Length_36 months	0.037504861
## Loan.Length_60 months	-0.037504861
## Loan.Purpose_car	NA
## Loan.Purpose_credit_card	NA
## Loan.Purpose_debt_consolidation	NA
## Loan.Purpose_educational	NA
## Loan.Purpose_home_improvement	NA
## Loan.Purpose_house	NA
## Loan.Purpose_major_purchase	NA
## Loan.Purpose_medical	NA
## Loan.Purpose_moving	NA
## Loan.Purpose_other	NA
## Loan.Purpose_renewable_energy	NA
## Loan.Purpose_small_business	NA
## Loan.Purpose_vacation	NA
## Loan.Purpose_wedding	NA
## Loan.Purpose_NA	-0.004862448
## Home.Ownership_MORTGAGE	0.032013969
## Home.Ownership_OTHER	-0.009731592
## Home.Ownership_OWN	-0.005283192
## Home.Ownership_RENT	-0.028322902
## Employment.Length_< 1 year	-0.076997865
## Employment.Length_0	-0.041341374
## Employment.Length_1 year	-0.063446275
## Employment.Length_2 years	-0.075269314
## Employment.Length_3 years	-0.072531378
## Employment.Length_4 years	-0.063878139
## Employment.Length_5 years	-0.068090583
## Employment.Length_6 years	-0.058085838
## Employment.Length_7 years	1.000000000
## Employment.Length_8 years	-0.048181633
## Employment.Length_9 years	-0.039790472
## Employment.Length_10+ years	-0.135192663
##	Employment.Length_8 years
## ID	NA
## Amount.Requested	0.008558836
## Amount.Funded.By.Investors	NA
## Monthly.Income	0.011149436
## Open.CREDIT.Lines	NA

## Revolving.CREDIT.Balance	NA
## Inquiries.in.the.Last.6.Months	0.010316480
## Loan.Length_36 months	-0.013877248
## Loan.Length_60 months	0.013877248
## Loan.Purpose_car	NA
## Loan.Purpose_credit_card	NA
## Loan.Purpose_debt_consolidation	NA
## Loan.Purpose_educational	NA
## Loan.Purpose_home_improvement	NA
## Loan.Purpose_house	NA
## Loan.Purpose_major_purchase	NA
## Loan.Purpose_medical	NA
## Loan.Purpose_moving	NA
## Loan.Purpose_other	NA
## Loan.Purpose_renewable_energy	NA
## Loan.Purpose_small_business	NA
## Loan.Purpose_vacation	NA
## Loan.Purpose_wedding	NA
## Loan.Purpose_NA	-0.004543294
## Home.Ownership_MORTGAGE	0.061264694
## Home.Ownership_OTHER	-0.009092844
## Home.Ownership_OWEN	-0.013435868
## Home.Ownership_RENT	-0.053196523
## Employment.Length_< 1 year	-0.071943991
## Employment.Length_0	-0.038627868
## Employment.Length_1 year	-0.059281880
## Employment.Length_2 years	-0.070328896
## Employment.Length_3 years	-0.067770668
## Employment.Length_4 years	-0.059685398
## Employment.Length_5 years	-0.063621352
## Employment.Length_6 years	-0.054273284
## Employment.Length_7 years	-0.048181633
## Employment.Length_8 years	1.000000000
## Employment.Length_9 years	-0.037178762
## Employment.Length_10+ years	-0.126319083
## Employment.Length_9 years	
## ID	NA
## Amount.Requested	0.050315258
## Amount.Funded.By.Investors	NA
## Monthly.Income	-0.002684880
## Open.CREDIT.Lines	NA
## Revolving.CREDIT.Balance	NA
## Inquiries.in.the.Last.6.Months	-0.016358220
## Loan.Length_36 months	-0.025150714
## Loan.Length_60 months	0.025150714
## Loan.Purpose_car	NA
## Loan.Purpose_credit_card	NA
## Loan.Purpose_debt_consolidation	NA
## Loan.Purpose_educational	NA
## Loan.Purpose_home_improvement	NA
## Loan.Purpose_house	NA
## Loan.Purpose_major_purchase	NA
## Loan.Purpose_medical	NA
## Loan.Purpose_moving	NA

## Loan.Purpose_other	NA
## Loan.Purpose_renewable_energy	NA
## Loan.Purpose_small_business	NA
## Loan.Purpose_vacation	NA
## Loan.Purpose_wedding	NA
## Loan.Purpose_NA	-0.003752048
## Home.Ownership_MORTGAGE	0.021317596
## Home.Ownership_OTHER	-0.007509263
## Home.Ownership_OWN	-0.012566518
## Home.Ownership_RENT	-0.013810099
## Employment.Length_< 1 year	-0.059414452
## Employment.Length_0	-0.031900560
## Employment.Length_1 year	-0.048957535
## Employment.Length_2 years	-0.058080637
## Employment.Length_3 years	-0.055967942
## Employment.Length_4 years	-0.049290777
## Employment.Length_5 years	-0.052541258
## Employment.Length_6 years	-0.044821220
## Employment.Length_7 years	-0.039790472
## Employment.Length_8 years	-0.037178762
## Employment.Length_9 years	1.000000000
## Employment.Length_10+ years	-0.104319749
##	Employment.Length_10+ years
## ID	NA
## Amount.Requested	0.148552254
## Amount.Funded.By.Investors	NA
## Monthly.Income	0.093489518
## Open.CREDIT.Lines	NA
## Revolving.CREDIT.Balance	NA
## Inquiries.in.the.Last.6.Months	-0.010197223
## Loan.Length_36 months	-0.047243430
## Loan.Length_60 months	0.047243430
## Loan.Purpose_car	NA
## Loan.Purpose_credit_card	NA
## Loan.Purpose_debt_consolidation	NA
## Loan.Purpose_educational	NA
## Loan.Purpose_home_improvement	NA
## Loan.Purpose_house	NA
## Loan.Purpose_major_purchase	NA
## Loan.Purpose_medical	NA
## Loan.Purpose_moving	NA
## Loan.Purpose_other	NA
## Loan.Purpose_renewable_energy	NA
## Loan.Purpose_small_business	NA
## Loan.Purpose_vacation	NA
## Loan.Purpose_wedding	NA
## Loan.Purpose_NA	-0.012748012
## Home.Ownership_MORTGAGE	0.190211445
## Home.Ownership_OTHER	-0.001139398
## Home.Ownership_OWN	-0.024130156
## Home.Ownership_RENT	-0.177130850
## Employment.Length_< 1 year	-0.201867372
## Employment.Length_0	-0.108385790
## Employment.Length_1 year	-0.166338804

```
corrplot.mixed(cr)
```




```
##
## Call:
## lm(formula = Amount.Requested ~ Monthly.Income, data = dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -51970  -5247  -1366   4484  22721
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   8.262e+03  2.668e+02  30.97  <2e-16 ***
## Monthly.Income 7.417e-01  3.798e-02  19.53  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7236 on 2180 degrees of freedom
## Multiple R-squared:  0.1489, Adjusted R-squared:  0.1485
## F-statistic: 381.3 on 1 and 2180 DF, p-value: < 2.2e-16
```

```
model2 = lm(Amount.Requested ~ Monthly.Income + `Loan.Length_36 months`, data = dataset)
summary(model2)
```

```
##
## Call:
## lm(formula = Amount.Requested ~ Monthly.Income + `Loan.Length_36 months`,
##      data = dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -47346  -4533   -905   3854  23429
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.426e+04  3.719e+02  38.34  <2e-16 ***
## Monthly.Income    6.904e-01  3.465e-02  19.93  <2e-16 ***
## `Loan.Length_36 months` -7.290e+03  3.424e+02 -21.29  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6585 on 2179 degrees of freedom
## Multiple R-squared:  0.2954, Adjusted R-squared:  0.2948
## F-statistic: 456.9 on 2 and 2179 DF, p-value: < 2.2e-16
```

```
vif(model2)
```

```
##           Monthly.Income `Loan.Length_36 months`
##           1.004847           1.004847
```

```
## model3 <- lm(Amount.Requested ~ Monthly.Income + `Loan.Length_36 months` + `Loan.Length_60 months`, data = dataset)
## summary(model3)
## vif(model3) ~ Not getting vif since Loan.Length_36 months` + `Loan.Length_60 months` are highly corelated so i am dropping this

model4 <- lm(Amount.Requested ~ Monthly.Income + `Loan.Length_36 months` + Home.Ownership_MORTGAGE, data = dataset)
summary(model4)
```

```
##
## Call:
## lm(formula = Amount.Requested ~ Monthly.Income + `Loan.Length_36 months` + Home.Ownership_MORTGAGE, data = dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -46641  -4494   -836    3771   23771
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    13988.1487    386.7158   36.172 <2e-16 ***
## Monthly.Income      0.6718      0.0354   18.979 <2e-16 ***
## `Loan.Length_36 months` -7237.8990    342.6513  -21.123 <2e-16 ***
## Home.Ownership_MORTGAGE  726.4915    289.6573    2.508  0.0122 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6577 on 2178 degrees of freedom
## Multiple R-squared:  0.2975, Adjusted R-squared:  0.2965
## F-statistic: 307.4 on 3 and 2178 DF, p-value: < 2.2e-16
```

```
vif(model4)
```

```
##           Monthly.Income `Loan.Length_36 months` Home.Ownership_MORTGAGE
##           1.051331           1.008595           1.052066
```

```
model5 <- lm(Amount.Requested ~ Monthly.Income + `Loan.Length_36 months` + Home.Ownership_MORTGAGE + Inquiries.in.the.Last.6.Months, data = dataset)
summary(model5)
```

```
##
## Call:
## lm(formula = Amount.Requested ~ Monthly.Income + `Loan.Length_36 months` +
##      Home.Ownership_MORTGAGE + Inquiries.in.the.Last.6.Months,
##      data = dataset)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -45845   -4471    -887    3820   23776
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.425e+04  3.991e+02   35.70 < 2e-16 ***
## Monthly.Income      6.736e-01  3.536e-02   19.05 < 2e-16 ***
## `Loan.Length_36 months` -7.271e+03  3.424e+02  -21.23 < 2e-16 ***
## Home.Ownership_MORTGAGE      7.801e+02  2.900e+02    2.69 0.00721 **
## Inquiries.in.the.Last.6.Months -2.993e+02  1.160e+02   -2.58 0.00993 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6569 on 2177 degrees of freedom
## Multiple R-squared:  0.2996, Adjusted R-squared:  0.2983
## F-statistic: 232.8 on 4 and 2177 DF, p-value: < 2.2e-16
```

```
vif(model5)
```

```
##              Monthly.Income      `Loan.Length_36 months`
##              1.051770              1.009990
##      Home.Ownership_MORTGAGE Inquiries.in.the.Last.6.Months
##              1.057485              1.008430
```